Hierarchical percolation of travel time recovers the provincial structure of the Roman road network

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Abstract

We apply hierarchical agglomerative clustering (HAC) on travel-time distances – mathematically equivalent to a percolation sweep over the network's edges – to the Roman road network, and ask whether the clusters that emerge at different travel-time thresholds recover the Roman provinces. We first validate the method on the exact network and travel-time model used in a recent large-scale reconstruction of the Roman road system, recovering the provincial partition with an Adjusted Rand Index (ARI) of 0.70 (bootstrap 95% CI [0.68, 0.71]) at a threshold of ∼77 hours (∼3.2 days) of travel. We then apply the same method to our own independently cleaned network under a simpler, constant-speed travel-time model, where the provincial alignment is weaker (ARI = 0.43, 95% CI [0.43, 0.45]) and peaks at a shorter travel-time scale (∼25–32 hours). Together these results indicate that administrative provincial boundaries of the Roman Empire left a recoverable, threshold-dependent signature in the connectivity structure of its road network, and that the strength of that signature is sensitive to the travel-time model used to weight the network.

1 Introduction

Introduction placeholder – to be completed.

2 Methods

2.1 Data

Road network. The primary network is built from the raw Itiner-e linework [de Soto et al., 2025]. Near-miss junctions and duplicate endpoints are snapped, the linework is noded, and degree-2 chains are simplified, giving a weighted graph of 8,460 nodes and 12,849 edges across 12 connected components (see `data_cleaning/` in the accompanying repository). Edge weights are travel times computed from edge length at a constant walking speed of 2 km/h.

Validation network. To check that our results are not an artefact of our own cleaning pipeline or travel-time model, we separately replicate the analysis on the exact Itiner-e export and endpoint-noded topology used by Pažout et al. [2026] (10,506 nodes, 14,770 edges, 13 connected components, including ferry links across the Bosporus and the Dardanelles), with travel time computed from Tobler's hiking function [Tobler, 1993] applied to slope and geodesic edge length. This self-contained replication is kept in `results/HAC/paper_replication/` and does not otherwise modify the primary analysis.

Province boundaries. Both networks are overlaid with the Ancient World Mapping Center's "Roman Empire AD 200" province polygons, corrected by A. Pažout [Ancient World Mapping Center, 2023] (57 provinces). Each edge is assigned to the province containing its midpoint, falling back to the province with the longest geometric overlap when the midpoint falls outside every polygon; nodes inherit the province of their incident within-province edges. Edges that cross a provincial boundary, or that cannot be assigned to any province, are excluded from the alignment scoring described below.

2.2 Hierarchical agglomerative clustering and percolation

Following the general approach of Arcaute et al. [2016], who use a percolation sweep over distance thresholds between street intersections to recover the hierarchical regional and urban structure of Britain, we apply an equivalent percolation sweep to travel time on the Roman road network itself, rather than to Euclidean distance between unconnected points.

Within each connected component, we compute pairwise travel-time distances between nodes and perform hierarchical agglomerative clustering (HAC) using single linkage. Because single linkage merges clusters at the weight of the minimum edge connecting them, cutting the resulting dendrogram at a travel-time threshold t reproduces exactly the connected components of the network with all edges of weight greater than t removed – i.e. HAC under single linkage is mathematically equivalent to a percolation sweep over the network's edges, computed once per component rather than independently at every threshold.

Sweeping t across the full range of merge heights gives, at each threshold: the number of clusters, and the size of the largest cluster relative to the network (the percolation curve). We additionally fit a power law to the percolation curve over its emergent region.

Alignment with provinces. At each threshold t, we compare the resulting cluster labels to the province labels of the nodes with a known province assignment, using the Adjusted Rand Index [ARI; Hubert and Arabie, 1985] and Normalized Mutual Information [NMI; Danon et al., 2005]. ARI ranges from 0 (random partition) to 1 (perfect agreement, up to cluster relabelling); NMI ranges from 0 to 1 and does not penalize a finer partition as harshly as ARI. We search exhaustively over every distinct merge height for the threshold that maximizes ARI against the provinces, and estimate a 95% confidence interval on that best ARI (and on the threshold it occurs at) by bootstrap resampling nodes with replacement and re-selecting, for each resample, the best-scoring threshold among those closest to the full-data optimum.

3 Results

3.1 Validation on the manuscript's network (first result)

Figure 1 shows the percolation curve and snapshot cluster maps for the validation network of Pažout et al. [2026], with Tobler-function travel times and AWMC provinces (10,430 of 10,506 nodes have a known province).

Table 1 reports ARI and NMI at six representative travel-time snapshots. Alignment with the provinces rises from essentially zero at 6–12 hours, peaks around 1–3 days, and falls off again as clusters grow beyond provincial scale.

An exhaustive search over every merge height finds a single, sharp peak at $t = 77.1$ hours ($\sim$3.2 days), with ARI $= 0.695$ (95% bootstrap CI $[0.681, 0.710]$). The peak is stable: every bootstrap resample of the node set selects the same 77-hour merge height as optimal. At this threshold the network's emergent regions correspond closely to the Roman provincial map.

Table 1: Province alignment at fixed travel-time snapshots, validation network (Pažout et al. [2026], Tobler-function travel time; $n = 10{,}430$ nodes with known province).

Snapshot	Clusters	ARI	NMI
6 h	2,809	0.039	0.640
12 h	1,376	0.092	0.683
24 h	514	0.201	0.734
2 d	154	0.516	0.803
7 d	22	0.493	0.778
12 d	15	0.308	0.689

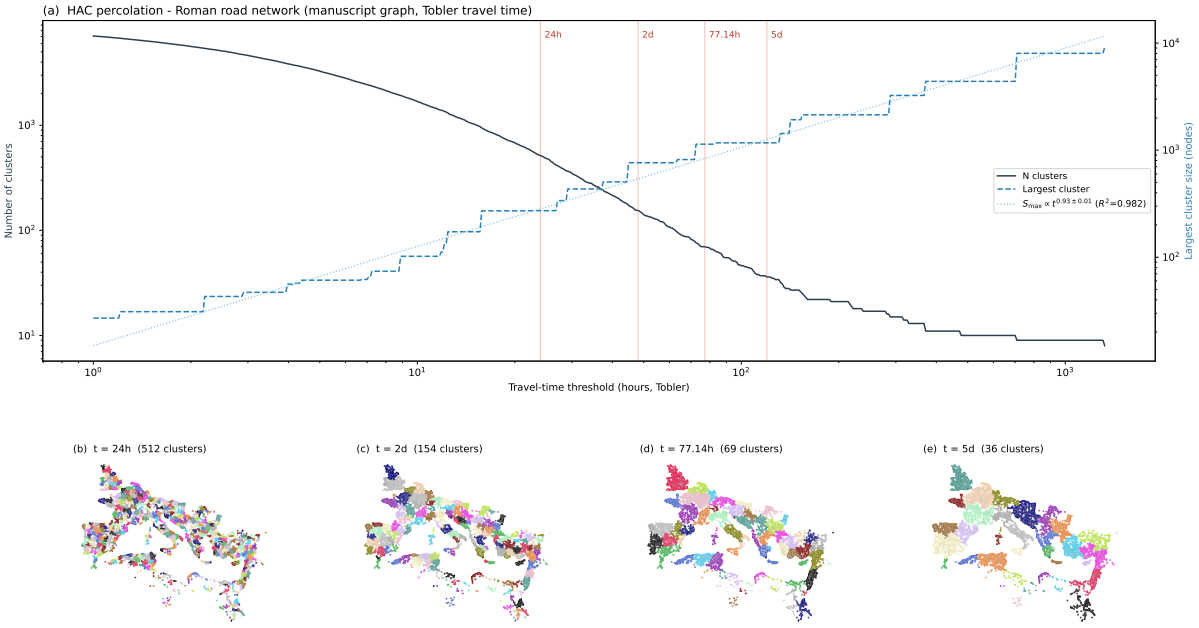


Figure 1: Percolation curve and snapshot cluster maps for the validation network (`results/HAC/paper_replication/maps/hac_percolation_combined.pdf`).

3.2 Percolation structure of the itinereX network (primary result)

Figure 2 shows the same analysis applied to our own independently cleaned network, with constant-speed (2 km/h) travel times (8,411 of 8,452 clustered nodes have a known province, across 56 represented provinces).

Table 2 reports the same six snapshots. As in Section 3.1, alignment is weak at sub-day thresholds and improves markedly by 24 hours, but here it falls away again by 2 days rather than continuing to improve through the 1–3 day range.

An exhaustive search over every merge height finds the best alignment at $t = 31.8$ hours, with ARI $= 0.433$ (NMI $= 0.763$, 291 clusters), 95% bootstrap CI on ARI of $[0.426, 0.448]$. Unlike the single sharp peak found in Section 3.1, the alignment curve here has two nearby local maxima of comparable height, around 25 and 32 hours; bootstrap resamples select between them, giving a wider 95% CI on the optimal threshold of $[24.8, 31.8]$ hours. The best achievable alignment with the provinces (ARI ≈ 0.43) is substantially weaker than the ≈ 0.70 recovered on the validation network of Section 3.1.

Table 2: Province alignment at fixed travel-time snapshots, primary itinereX network (constant 2 km/h travel time; $n = 8{,}411$ nodes with known province).

Snapshot	Clusters	ARI	NMI
6 h	4,455	0.029	0.615
12 h	2,464	0.135	0.666
24 h	652	0.418	0.753
2 d	76	0.048	0.395
7 d	8	0.018	0.209
12 d	7	0.017	0.199

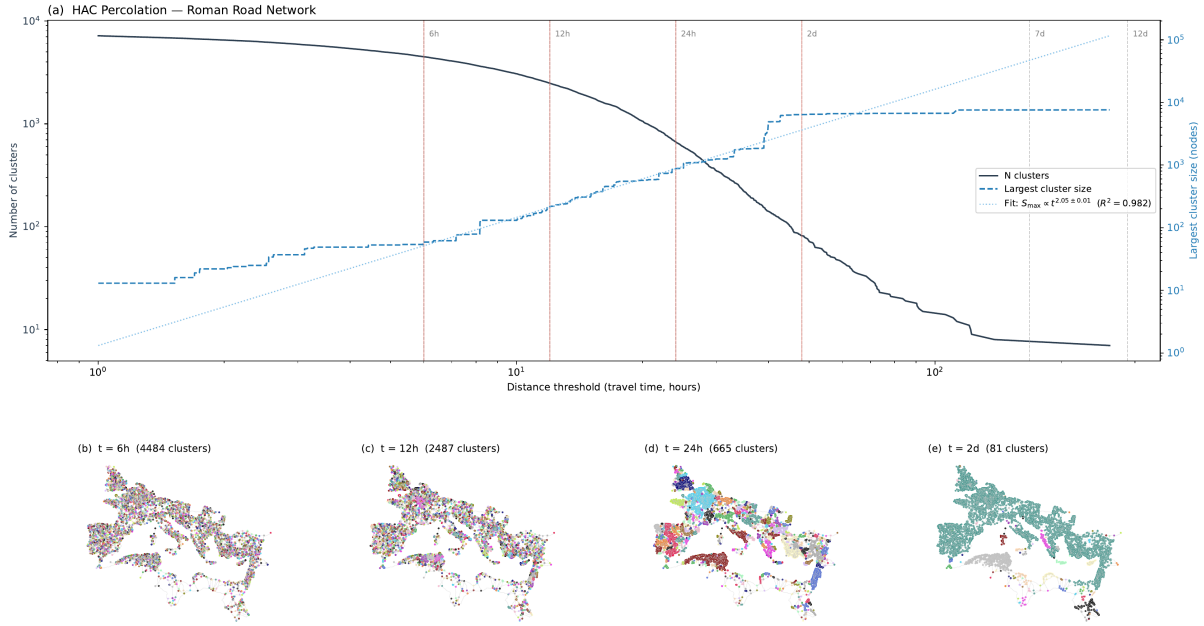


Figure 2: Percolation curve and snapshot cluster maps for the primary itinereX network (`results/HAC/maps/hac_percolation_combined.pdf`).

4 Discussion

Discussion placeholder – to be completed. Note for later: the gap between the validation network's ARI (≈0.70) and the primary network's ARI (≈0.43) is a candidate finding worth developing – it may reflect the travel-time model (Tobler hiking function vs. constant speed) rather than the underlying network topology, which could be tested using the spatial/temporal null models in `results/Null-models/`.

Data accessibility

All code and data underlying this analysis are available in the accompanying repository, under `results/HAC/` (primary analysis) and `results/HAC/paper_replication/` (validation analysis).

Authors' contributions

To be completed.

Competing interests

The authors declare no competing interests.

Funding

To be completed.

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